EE3110: Probability Foundations for EE

Marks: 40 | Final Exam, Nov 11, 2024 | Time: 3 Hours

Two A4 handwritten sheets allowed. All answers should be justified with clear reasons. State any reasonable assumptions you make. Examples cannot be used as a proof for a claim. They can be used to disprove a claim or to illustrate a particular case.

1. (8 marks)

   (a) (2 marks) Suppose X and Y are jointly Gaussian and their mean is zero, variance is 1 and correlation coefficient is 0.5. What is the distribution of $U = X + Y$? What are the mean and variance of U?

   (b) (2 marks) Suppose X has a moment generating function $M_X(s) = \frac{1}{3}\exp(s+s^2) + \frac{2}{3}\exp(-s+s^2)$ for all s. Find the expectation of X.

   (c) (2 marks) A random variable X has density $e^{-x}, x > 0$. Let $Y = \lfloor X \rfloor$, where $\lfloor x \rfloor$ denotes the floor function of x, or the greatest integer $\leq x$. Find $E[Y]$.
   Hint: $E[Y] = P(Y \geq 1) + P(Y \geq 2) + \cdots$

   (d) (2 marks) For a positive integer n, let X_n be a discrete random variable with $P(X_n = 0) = 1 - 1/n$ and $P(X_n = n) = 1/n$. Compute $E[X_n]$ and $\lim_{n\to\infty} P(X_n = 0)$.

2. (4 marks) Consider the sample space $\Omega = [0, 1]$ and a probability P on Ω such that for any interval $A \subset [0, 1]$. $P(A)$ is the length of A. Define a RV X on this Ω as follows: for positive integers $k \geq 1$, $X(\omega) = k$ if $2^{-k} < \omega \leq 2^{-(k-1)}$. Let Y be another random variable such that for any positive integer $x \geq 1$, $f_{Y|X}(y|x)$ is $\lambda_x \exp(-\lambda_x\, y)$ for $y \geq 0$, where $\lambda_x = 2^x$, and $f_{Y|X}(y|x)$ is 0 for $y < 0$.

   (a) (2 mark) Find $E[X|X \geq 2]$.

   (b) (2 marks) Find $E[Y]$.

3. (4 marks) Let $X_1, X_2, \ldots$ be independent random variables with the same mean μ and the same variance σ^2. Derive the following result: $P(|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu| > a) \leq \frac{\sigma^2}{na^2}$ for any $a > 0$. State any result that you use along the way before using it.

4. (4 marks) m balls are thrown into $2n$ bins with each ball independently and uniformly likely to go into any bin. Let a random variable X_i denote the number of balls in the i-th bin for $i = 1, 2, \ldots, 2n$. Define the indicator random variable $I(X_i = 0) = 1$ if $X_i = 0$ and 0 otherwise. Let a random variable $Y = \sum_{i=1}^{2n} I(X_i = 0)$ denote the number of empty bins. Find $E[Y]$ and $E[Y|X_1 + \cdots + X_n = 0]$.

5. (4 marks) Let random variables X and Y be jointly Gaussian with means $\mu_X = \mu_Y = 0$, variances $\sigma_X^2 = \sigma_Y^2 = 1$ and correlation coefficient ρ. Let $T = X\cos\theta + Y\sin\theta + 1$ and $W = X\sin\theta - Y\cos\theta - 1$. Find $P(T > 1|W = 0)$ in terms of ρ, θ and Φ_Z (CDF of the standard normal or Gaussian distribution $\mathcal{N}(0, 1)$).

6. (8 marks) Let X and Y be independent exponential random variables with parameters 1 and 3, respectively. Define $Z = \min(X, Y)$ and $V = \max(X, Y)$.

   (a) (2 marks) Find $P(V > 1)$.

   (b) (3 mark) Find $E[ZV]$ and $E[Z|V = X]$.

   (c) (3 marks) Let g be a function that minimizes $E[(Z - g(X))^2]$. Find the function $g(x)$ for $x \geq 0$ as an explicit form.

Hint: Try to use apply a result from Tutorial 4 for part (b). Before using a result from lectures and tutorials you must state the result clearly.

Please turn over...

7. (8 marks) You invest Rs. λT in shares of Company A at its current value of Rs. a_0 per share, and Rs. $(1-\lambda)T$ in shares of Company B at its current value b_0 per share, $0 \leq \lambda \leq 1$. The value per share of Company A after one month is modelled as $G_A a_0$, where $G_A \sim$ Uniform$[0, 4]$ is a continuous random variable. The value per share of Company B after one month is $G_B b_0$, where $G_B \sim$ Uniform$[0, 2]$ independent of G_A. The invested total amount T becomes $G_M T$ after one month, where G_M is a random variable. T, λ, a_0 and b_0 are non-random values.

(a) (3 marks) Write down an expression for G_M in terms of λ, G_A and G_B. Find $E[G_M]$ in terms of λ. What value of λ maximizes $E[G_M]$?

(b) (5 marks) The probability of losing half or more of your money is $P(G_M \leq 1/2)$. What value of λ minimizes this probability?

END

3. (4 marks) Let $X_1, X_2, \ldots$ be independent random variables with the same mean μ and the same variance σ^2. Derive the following result: $P(|\frac{1}{n}\sum_{i=1}^{n} X_i - \mu| > a) \leq \frac{\sigma^2}{na^2}$ for any $a > 0$. State any result that you use along the way before using it.

$$\bar{x} = \frac{1}{n}\left(\sum_{i=1}^{n} x_i\right)$$

$$\text{var}(\bar{x}) = \text{var}\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) = \frac{1}{n^2} \cdot n\sigma^2 = \sigma^2/n$$

$$P\left((\bar{x}-\mu)^2 > a^2\right) \leq \frac{E\left[(\bar{x}-\mu)^2\right]}{a^2} \rightarrow (\text{markov's inequality})$$

$$\leq \frac{\text{var}(\bar{x})}{a^2}$$

$$\leq \frac{\sigma^2}{na^2}$$

4. (4 marks) m balls are thrown into $2n$ bins with each ball independently and uniformly likely to go into any bin. Let a random variable X_i denote the number of balls in the i-th bin for $i = 1, 2, \ldots, 2n$. Define the indicator random variable $I(X_i = 0) = 1$ if $X_i = 0$ and 0 otherwise. Let a random variable $Y = \sum_{i=1}^{2n} I(X_i = 0)$ denote the number of empty bins. Find $E[Y]$ and $E[Y|X_1 + \cdots + X_n = 0]$.

First n-bins empty